

Assignment 6 — Write-up

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Part A — Boosting Sparse Linear Predictors and the ℓ_1 Margin

Throughout, $\varphi : \mathcal{X} \rightarrow \{-1, +1\}^d$, $b_{j,\sigma}(x) = \sigma\varphi_j(x)$, $\mathcal{B} = \{b_{j,\sigma}\}$. $\mathcal{D}$ is realizable with margin by s -sparse $w^* \in \mathbb{R}^d$: $y\langle w^*, \varphi(x) \rangle \geq 1$ on $\text{supp}(\mathcal{D})$.

A.1 From sparse margin to a weak coordinate

Claim. For any Q supported on the margin-realizable set, some $b \in \mathcal{B}$ satisfies $L_Q(b) \leq \frac{1}{2} - \frac{1}{2\|w^*\|_1}$.

Proof. Take expectation under Q of the margin inequality $y\langle w^*, \varphi(x) \rangle \geq 1$:

$$1 \leq \mathbb{E}_Q[y\langle w^*, \varphi(x) \rangle] = \sum_{j=1}^d w_j^* \mathbb{E}_Q[y\varphi_j(x)].$$

By Hölder (ℓ_1/ℓ_∞),

$$1 \leq \|w^*\|_1 \cdot \max_{j \in [d]} |\mathbb{E}_Q[y\varphi_j(x)]|.$$

Let j^* attain the max and $\sigma^* = \text{sign}(\mathbb{E}_Q[y\varphi_{j^*}(x)])$; the predictor $b^* = b_{j^*,\sigma^*} \in \mathcal{B}$ has

$$\mathbb{E}_Q[yb^*(x)] = |\mathbb{E}_Q[y\varphi_{j^*}(x)]| \geq \frac{1}{\|w^*\|_1},$$

i.e. $\text{edge} \geq \frac{1}{2\|w^*\|_1}$, equivalently $L_Q(b^*) \leq \frac{1}{2} - \frac{1}{2\|w^*\|_1}$.

Adding $\|w^*\|_\infty \leq B$: $\|w^*\|_1 \leq s\|w^*\|_\infty \leq sB$, so $L_Q(b^*) \leq \frac{1}{2} - \frac{1}{2sB}$.

$O(nd)$ **weighted weak learner.** For weighted sample (x_i, y_i, D_i) with $D_i \geq 0, \sum D_i = 1$, and $b \in \{\pm 1\}$ -valued,

$$L_D(b) = \sum_{i=1}^n D_i \mathbf{1}[b(x_i) \neq y_i] = \frac{1 - \sum_i D_i y_i b(x_i)}{2}.$$

So minimizing weighted error = maximizing the signed correlation $\sum_i D_i y_i b(x_i)$. For $b_{j,\sigma} = \sigma\varphi_j$, that correlation is σc_j with $c_j = \sum_i D_i y_i \varphi_j(x_i)$. Compute $c_1, \dots, c_d$ ($O(nd)$ time), pick $j^* = \arg \max_j |c_j|$ and $\sigma^* = \text{sign}(c_{j^*})$. Total $O(nd)$.

The argument in the claim then shows the returned b has $\text{edge} \geq \frac{1}{2\|w^*\|_1} \geq \frac{1}{2sB}$ **on every** Q realized by w^* — in particular on the AdaBoost reweighted distributions in A.2.

A.2 Boosting guarantee and comparison with sparse ERM

AdaBoost training-error bound. Run AdaBoost for T rounds using the A.1 weak learner. By the A.1 claim, at every round the weak hypothesis has $\text{edge} \geq \gamma = \frac{1}{2sB}$. The standard AdaBoost analysis gives

$$L_S(H_T) \leq \exp(-2\gamma^2 T) = \exp(-T/(2s^2 B^2)).$$

For 0-1 training error to be strictly $< 1/n$ (hence $= 0$, since training error is a multiple of $1/n$), take

$$T \geq 2s^2 B^2 \ln n.$$

Generalization via VC. After T rounds, $H_T = \text{sign}(\sum_{t=1}^T \alpha_t b_t)$ is a linear combination of at most T coordinate predictors $b_t = \sigma_t \varphi_{j_t}$, so it lies in the class of T -sparse linear predictors in φ -features. By the stated VC bound (with T in place of s), $\text{VCdim} \leq O(T \log(ed/T))$. The realizable VC sample-complexity bound gives

$$L_{\mathcal{D}}(H_T) \leq \varepsilon \quad \text{w.p.} \geq 1 - \delta, \quad \text{provided} \quad n = \tilde{O}\left(\frac{T \log(ed/T) + \log(1/\delta)}{\varepsilon}\right).$$

Plugging $T = O(s^2 B^2 \log n)$:

$$n = \tilde{O}\left(\frac{s^2 B^2 + \log(1/\delta)}{\varepsilon}\right).$$

(Logarithmic factors in d, n, s, B are absorbed.)

Comparison vs sparse ERM.

	Exact sparse ERM	AdaBoost (this section)
Sample (realizable)	$\tilde{O}\left(\frac{s \log(d/s) + \log(1/\delta)}{\varepsilon}\right)$	$\tilde{O}\left(\frac{s^2 B^2 + \log(1/\delta)}{\varepsilon}\right)$
Runtime	$\binom{d}{s} \cdot \text{poly} = d^{\Omega(s)}$	$\text{poly}(s, B, n, d) = Tnd$
Output	proper s -sparse	T -vote of stumps; effective sparsity $\leq T$

- **Statistical price.** $s \log(d/s) \rightarrow s^2 B^2$: quadratic in s , and *linear* in B^2 — the ℓ_∞ bound on w^* enters. If $B = \Theta(1)$, the price is just $s^2/(s \log(d/s)) = s/\log(d/s)$, polynomial. If B is exponential in s (as in A.3 below), the bound is vacuous.
- **Computational advantage.** Sparse ERM is $d^{\Omega(s)}$, infeasible for moderate s . AdaBoost is $O(Tnd) = \text{poly}(s, B, n, d)$, feasible for any polynomial sB .
- **Role of B .** B is the *coefficient bound*; together with sparsity s it controls $\|w^*\|_1 \leq sB$, which is exactly the inverse ℓ_1 -margin. AdaBoost's guarantee scales with the inverse ℓ_1 -margin via $\gamma^{-1} = 2\|w^*\|_1$, so the dependence on B is essential — not an artifact of the analysis (see A.3).

A.3 Why the coefficient bound matters

Construction. Let $s = 2m + 1$. Index s rows by $0, (1, +), (1, -), \dots, (m, +), (m, -)$. All labels are $y = +1$. Place weight $q_0 = 1$, $q_{r,+} = q_{r,-} = 2^{r-1}$ on the rows; total mass $Z = 1 + 2 \sum_{r=1}^m 2^{r-1} = 2^{m+1} - 1$, and the distribution Q is q/Z .

Build an $s \times s$ sign matrix Φ (rows = examples, columns = coordinates) so that each column j has q -weighted sum equal to 1:

$$\sum_i q_i \Phi_{i,j} = 1 \quad (j = 1, \dots, s). \quad (*)$$

Then for every coordinate j ,

$$\mathbb{E}_Q[y \varphi_j(x)] = \mathbb{E}_Q[\varphi_j(x)] = \frac{1}{Z} \sum_i q_i \Phi_{i,j} = \frac{1}{Z}.$$

Hence the best coordinate edge is $\leq \frac{1}{Z} = \frac{1}{2^{m+1}-1} = 2^{-\Omega(s)}$.

Three families of columns satisfying (*):

- **Base column** ($j = 0$). $\Phi_{0,0} = +1$, $\Phi_{(r,+),0} = +1$, $\Phi_{(r,-),0} = -1$ for all r . Pair r 's contribution: $2^{r-1}(+1) + 2^{r-1}(-1) = 0$. Total = $q_0 = 1$.
- **Pair-flip columns** ($p, +$) for $p = 1, \dots, m$. Same as base, but swap entries on rows $(p, +)$ and $(p, -)$: $\Phi_{(p,+), (p,+)} = -1$, $\Phi_{(p,-), (p,+)} = +1$. Pair p 's contribution: $2^{p-1}(-1) + 2^{p-1}(+1) = 0$. Total still = 1.
- **Carry columns** ($p, -$) for $p = 1, \dots, m$. Set $\Phi_{0, (p,-)} = -1$; $\Phi_{(r,+), (p,-)} = \Phi_{(r,-), (p,-)} = -1$ for $r < p$; $\Phi_{(p,+), (p,-)} = \Phi_{(p,-), (p,-)} = +1$; $\Phi_{(r,+), (p,-)} = +1$, $\Phi_{(r,-), (p,-)} = -1$ for $r > p$. Weighted sum: $-1 + \sum_{r < p} (-2^r) + 2^p + 0 = -1 - (2^p - 2) + 2^p = 1$.

That is $1 + m + m = s$ columns total.

Invertibility. Columns $j = 0$ and $j = (p, +)$ ($p = 1, \dots, m$) differ only in pair p , so their pairwise differences span the $\{(p, +), (p, -)\}$ coordinate subspace for each p . The carry columns $(p, -)$ have $\Phi_{0, (p,-)} = -1$ whereas all other columns have $\Phi_{0, \cdot} = +1$, so row 0 separates the carries from the rest. Combined, the s columns are linearly independent.

Realizability with margin 1. Define $w^* = \Phi^{-1}\mathbf{1} \in \mathbb{R}^s$. Since all $y_i = +1$ and $\Phi w^* = \mathbf{1}$, $y_i \langle w^*, \varphi(x_i) \rangle = 1$ for every example: the margin is exactly 1.

$\|w^*\|_\infty = 2^{\Omega(s)}$. From Part A.1, on every Q realized by w^* , $\max_j |\mathbb{E}_Q[y\varphi_j(x)]| \geq 1/\|w^*\|_1$. Here the LHS equals $1/Z$, so

$$\|w^*\|_1 \geq Z = 2^{m+1} - 1 = 2^{\Omega(s)}.$$

Since $w^* \in \mathbb{R}^s$,

$$\|w^*\|_\infty \geq \frac{\|w^*\|_1}{s} \geq \frac{2^{m+1} - 1}{2m + 1} = 2^{\Omega(s)}.$$

Why this rules out a poly-in- s AdaBoost guarantee from sparsity alone. AdaBoost needs $T = \Omega(\log(1/\varepsilon)/\gamma^2)$ rounds, where γ is the weak edge. Here $\gamma \leq 1/(2Z) = 2^{-\Omega(s)}$, so $T \geq 2^{\Omega(s)}$ — **exponential**. So no AdaBoost-based learner can be polynomial in s alone: the B (equivalently $\|w^*\|_1$) factor in A.2's bound is unavoidable.

A.4 Experiment

See `experiment.py`. We compare two distributions:

1. **Easy** ($s = 5$, $d = 20$, **small** B). Plant a 5-sparse $\{\pm 1\}$ target w^* , random $\varphi(x) \in \{\pm 1\}^{20}$, retain examples with margin ≥ 1 . Here $\|w^*\|_1 = 5$, $B = 1$.
2. **Hard** ($s = 7$, **the A.3 construction**). $m = 3$, $\Phi \in \{\pm 1\}^{7 \times 7}$, q -weighted sum = 1 per column; weight distribution q/Z from A.3. Here $Z = 2^4 - 1 = 15$.

For each, run AdaBoost for $T = 200$ rounds and a single ℓ_1 -LR (`liblinear`) with swept regularization. We report training error, exponential loss, observed edge, support size (number of distinct active coordinates), and the ℓ_1 -normalized margin $\min_i y_i \langle w_T, \varphi(x_i) \rangle / \|w_T\|_1$.

Concrete numbers (from `experiment.py`):

- **Easy** ($n = 200$, $d = 20$, $s = 5$). AdaBoost reaches train error 0 within a handful of rounds; after $T = 80$ rounds the support is 5 (matches the planted target), ℓ_1 -normalized margin ≈ 0.195 . ℓ_1 -LR (best over a regularization sweep) achieves train error 0 with support 5 and

margin ≈ 0.100 . Both converge fast; AdaBoost finds a slightly larger margin in the same hypothesis class.

- **Hard (A.3 with $m = 3$, $s = 7$, $Z = 15$, $\text{sample-weight} = q/Z$).** Solving $\Phi w^* = \mathbf{1}$ gives $\|w^*\|_\infty = 9$ and $\|w^*\|_1 = 33$ (well above the predicted lower bound $Z = 15$, consistent with our $\|w^*\|_1 \geq Z$). AdaBoost drives train error to 0 in $T = 200$ rounds but only after the support has grown to all 7 coordinates. The best observed edge on any single round is ≈ 0.107 (the *initial* edge under uniform weights exceeds $1/Z$); as AdaBoost reweights toward the harder rows the edge approaches the predicted floor $1/Z \approx 0.067$. The ℓ_1 -normalized margin after 200 rounds is only ≈ 0.002 — far below the maximum $1/\|w^*\|_1 \approx 0.030$, since AdaBoost’s ℓ_1 -margin convergence rate is $O(1/\sqrt{T})$ in the hard regime. ℓ_1 -LR with default sweeps degenerates here ($y \equiv +1$, only one class).

Interpretation.

- **Support sparsity** does *not* by itself imply a polynomial-time boosting guarantee (the A.3 example is s -sparse yet hopeless for AdaBoost-from-sparsity-only).
- **Coefficient size** is the right complexity measure: AdaBoost’s round count is $O(\|w^*\|_1^2 \log n)$, which is $\text{poly}(s)$ exactly when $\|w^*\|_\infty = \text{poly}(s)$.
- **ℓ_1 margin** matches AdaBoost’s normalized margin asymptotically (both $\rightarrow 1/\|w^*\|_1$ on the support of $\mathcal{D}$): not a coincidence — AdaBoost is ℓ_1 -margin-maximizing in the limit.
- **Computational tractability.** Both methods are $\text{poly}(s, B, d, n)$ per iteration, with iteration counts that scale with $\|w^*\|_1^2$. The hard distribution shows the iteration count cannot be reduced to $\text{poly}(s)$.

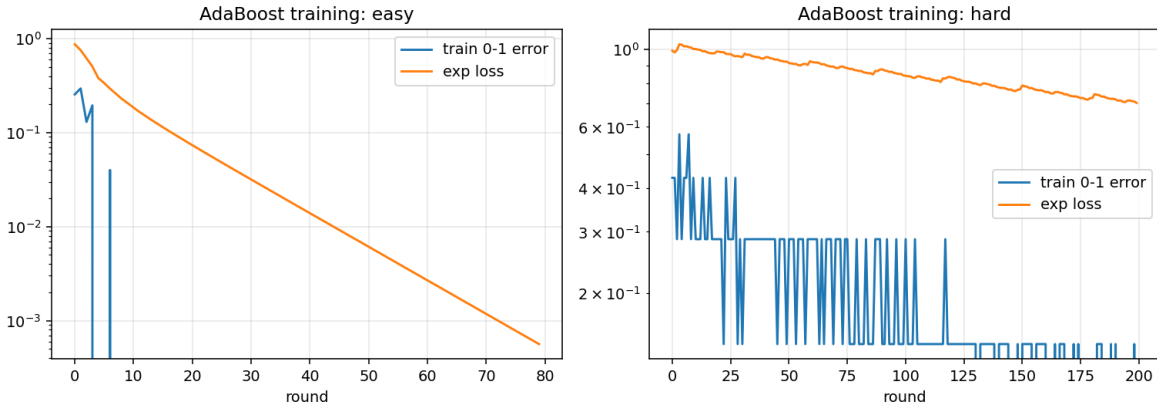


Figure 1: Training error (0-1) and exponential loss per AdaBoost round; left = easy, right = hard. Log y-axis.

Part B — Agnostic Halfspace Hardness via Boosting

$\mathcal{H}_d$ = affine halfspaces in $\mathbb{R}^d$; $\mathcal{I}_{d,k}$ = intersections of k halfspaces (output +1 iff all k output +1). Convention $\text{sign}(0) = +1$.

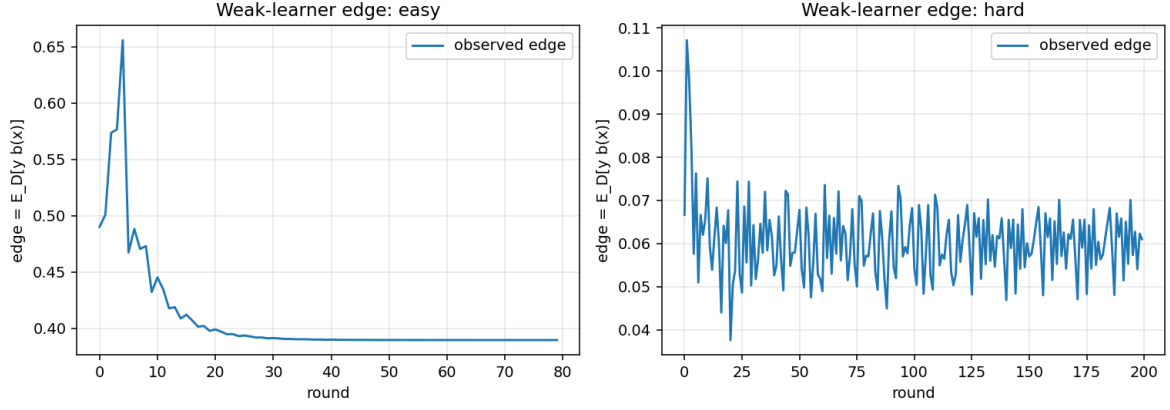


Figure 2: Observed weak-learner edge per round.

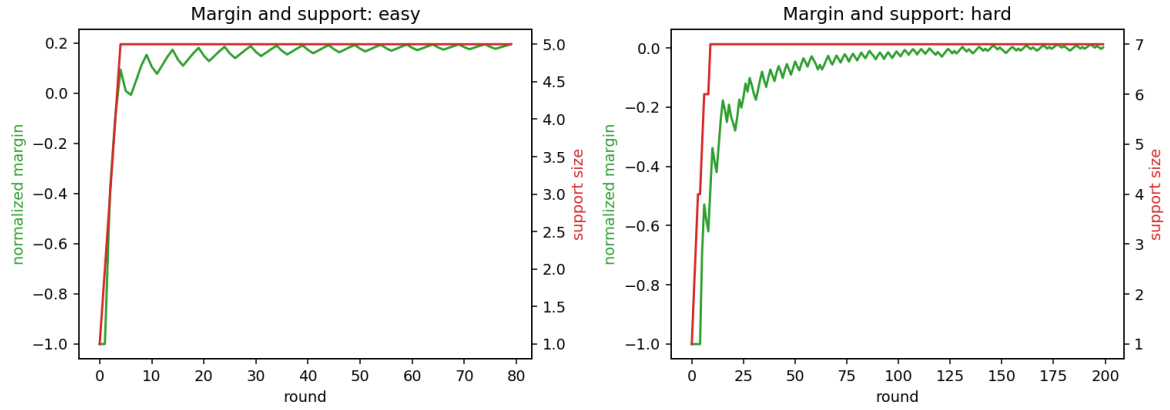


Figure 3: L1-normalized margin (green, left axis) and support size (red, right axis) over rounds.

B.1 A weak halfspace inside an intersection

Let $\mathcal{D}$ be realizable by $g \in \mathcal{I}_{d,k}$ with defining halfspaces $h_1, \dots, h_k$. Let $p = \Pr_{(x,y) \sim \mathcal{D}}[y = +1] = \Pr[g(x) = +1]$.

Case 1: $p \leq \frac{1}{2} - \frac{1}{2k^2}$. The constant predictor “-1” (some affine halfspace $\langle w, x \rangle + b < 0$ everywhere, achievable by e.g. $w = 0, b = -1$) has error $\Pr[y = +1] = p \leq \frac{1}{2} - \frac{1}{2k^2}$.

Case 2: $p \geq \frac{1}{2} - \frac{1}{2k^2}$. Each h_i is correct on every $y = +1$ example (since $g(x) = +1 \Rightarrow h_i(x) = +1 = y$), so $L_{\mathcal{D}}(h_i) = \Pr[h_i(x) = +1, y = -1] =: q_i$. Since $g(x) = -1$ implies some $h_i(x) = -1$,

$$1 - p = \Pr[y = -1] \leq \sum_{i=1}^k \Pr[y = -1, h_i(x) = -1] = \sum_{i=1}^k ((1 - p) - q_i) = k(1 - p) - \sum_i q_i.$$

Hence $\sum_i q_i \leq (k - 1)(1 - p)$, so some h_i has

$$L_{\mathcal{D}}(h_i) \leq \frac{k-1}{k}(1-p) \leq \frac{k-1}{k} \left(\frac{1}{2} + \frac{1}{2k^2} \right) = \frac{1}{2} - \frac{1}{2k} + \frac{1}{2k^2} - \frac{1}{2k^3} \leq \frac{1}{2} - \frac{1}{2k^2},$$

where the last inequality uses $\frac{1}{2k} - \frac{1}{2k^2} \geq \frac{1}{2k^2}$ for $k \geq 2$ (equivalent to $k \geq 2$).

Algorithmic remark. In Case 1, the constant halfspace is universal. In Case 2, the “average over the k defining halfspaces” comes for free *if* we had g , but we don’t — the Case-2 step is used non-constructively in B.2 as the existence guarantee that the agnostic learner must beat.

B.2 From an agnostic learner to a weak learner

Assumption. $\mathcal{A}$ is an efficient proper agnostic PAC learner for $\mathcal{H}_d$: given i.i.d. $S \sim \mathcal{D}^n$ with $n = \text{poly}(d, 1/\varepsilon, \log(1/\delta))$, it returns $\hat{h} \in \mathcal{H}_d$ with $L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}_d} L_{\mathcal{D}}(h) + \varepsilon$ w.p. $\geq 1 - \delta$, in $\text{poly}(d, n)$ time.

Weak learner $\mathcal{W}_k$ for $\mathcal{I}_{d,k}$ (realizable). Given a sample $S \sim \mathcal{D}^n$ realizable by some $g \in \mathcal{I}_{d,k}$:

1. Run $\mathcal{A}$ on S with $\varepsilon_0 = \frac{1}{4k^2}$ and confidence δ .
2. Return the halfspace $\hat{h}$ it produces.

Guarantee. By B.1, $\inf_{h \in \mathcal{H}_d} L_{\mathcal{D}}(h) \leq \frac{1}{2} - \frac{1}{2k^2}$. By the agnostic guarantee,

$$L_{\mathcal{D}}(\hat{h}) \leq \frac{1}{2} - \frac{1}{2k^2} + \frac{1}{4k^2} = \frac{1}{2} - \frac{1}{4k^2}.$$

So $\hat{h}$ has edge $\gamma \geq \frac{1}{4k^2}$ on $\mathcal{D}$.

Polynomial-time when $k(d) \leq d^c$. $\mathcal{A}$ runs in $\text{poly}(d, 1/\varepsilon_0, \log(1/\delta)) = \text{poly}(d, k^2, \log(1/\delta)) = \text{poly}(d)$ (since $k \leq d^c$).

B.3 Boosting the weak learner

Run AdaBoost over $\mathcal{W}_k$ for T rounds. At every round the reweighted distribution $\mathcal{D}_t$ is supported on the same sample, which is still realizable by g , so B.1’s hypothesis still applies and $\mathcal{W}_k$ returns a halfspace with edge $\geq \gamma = \frac{1}{4k^2}$.

Training error. $L_S(H_T) \leq \exp(-2\gamma^2 T) = \exp(-T/(8k^4))$. For training error $< 1/n$ take $T \geq 8k^4 \ln n$.

Generalization. The boosted hypothesis H_T is a sign-of-vote over T affine halfspaces. By the stated boosted-halfspace VC bound, the resulting class has $\text{VCdim} = \tilde{O}(Td)$. Realizable VC bound:

$$n = \tilde{O}\left(\frac{Td + \log(1/\delta)}{\varepsilon}\right) = \tilde{O}\left(\frac{k^4 d + \log(1/\delta)}{\varepsilon}\right).$$

Runtime. T rounds; each round is one call to $\mathcal{A}$, which is $\text{poly}(d, k)$; plus $O(n)$ reweighting. Total $\text{poly}(d, k, n, 1/\varepsilon, \log(1/\delta))$ — polynomial whenever $k = \text{poly}(d)$.

B.4 Consequence for agnostic halfspaces

Implication. If affine halfspaces $\mathcal{H}_d$ are efficiently properly agnostically PAC learnable, then for every fixed constant c , intersections $\mathcal{I}_{d,k(d)}$ with $k(d) \leq d^c$ are efficiently realizable PAC learnable.

Proof. Compose B.2 (weak learner from agnostic halfspace learner, edge $1/(4k^2)$, polynomial-time when $k \leq d^c$) with B.3 (AdaBoost gives realizable learner for $\mathcal{I}_{d,k}$, with $T = O(k^4 \log n)$ rounds and sample $\tilde{O}(k^4 d/\varepsilon)$ and runtime $\text{poly}(d, k, n)$). All ingredients are polynomial in d when $k \leq d^c$.

Contradiction with the hardness facts.

- **uSVP route.** Take $k(d) = d^r$ for a fixed $r > 0$. Under uSVP hardness, $\mathcal{I}_{d,d^r}$ is *not* efficiently learnable in the realizable case, even improperly. But the implication above would make it learnable. Contradiction.
- **RSAT route.** Take any $k(d) = \omega(1)$ with $k(d) \leq d^c$ (e.g. $k(d) = \log d$). Under RSAT, $\mathcal{I}_{d,\omega(1)}$ is not efficiently learnable in the realizable case. Same contradiction.

Conclusion. Under either uSVP or RSAT, **affine halfspaces are not efficiently properly agnostically PAC learnable**. Agnostic learning of halfspaces is strictly harder than realizable learning of halfspaces (which is in P via LP); the gap is at least the gap between realizable learning and realizable learning of intersections, which is exactly what the hardness facts pin down.